

Membrane Air Separation

Pre-Lab Plan Unit Operations Lab # 5

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Group 3, Tuesday, PM section

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1. Experimental Objective (5 pts)

The goal of this experiment is to obtain overall mass transport coefficients for the oxygen and nitrogen moving through the separation unit by measuring the volumetric flowrate on either side of the membrane at varying operating pressures and tube side flowrates. These parameters will be varied by manipulating the pressure of the gas entering the system and the connection method (series vs. parallel) of the two separation modules.

2. Experimental

For this experiment, the gas permeation apparatus (shown below in **Figure 1**) is used to complete the separation of nitrogen and oxygen. The extra-dry compressed air feed can be turned on using the valves on the cylinder. The pressure entering the system is measured by a strain-gage transducer and can be read on the digital display (labelled #4). The oxygen level of the incoming stream can be read through the oxygen analyzer (#3).

The separation begins with a gaseous feed stream being fed into two Prism separator modules (#1) which contain the hollow fibers which act as non-porous semipermeable membranes. Since the fiber wall is more permeable to oxygen than nitrogen, the stream on the shell side of the module is more oxygen rich and the stream leaving the other side will be depleted in oxygen.

When connected in parallel, the permeate effluent from both modules are combined before passing through the oxygen analyzer (#3) and flow meter (#2). The non-permeate streams are also combined before the same analysis is conducted. In series, the shell-side flowrate and oxygen content flowing out of each module can be measured separately. The configuration (parallel vs. series) of the system can be controlled through the position of valves A and C (#6).

2.1 Apparatus (5 pts)

Figure 1

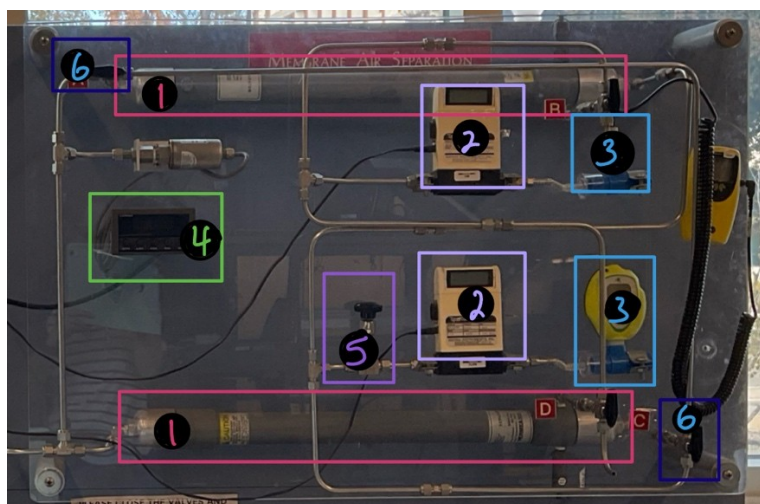


Table 1: Relates the color-coded labels to what part of the apparatus they are.

Number/Color	Apparatus Part
1 – Pink	Modules with membranes
2 – Light Purple	Mass flow meters
3 – Light Blue	Oxygen sensors
4 – Light Green	Pressure gauge
5 – Purple	Air flow control valve
6 – Dark Blue	Three-way valves

2.2 Materials and Supplies (5 pts)

Table 2: A list of the materials/supplies used for lab.

Material/Supplies	Purpose/Specifications
Dry Compressed Air	This is the air that will go through the separation process
Regulator	To regulate the inlet of the compressed air
Shutoff Valve	Attached as a safety mechanism to the compressed air

2.3 Experimental Plan (10 pts)

- Power on the apparatus and allow the system to warm up and calibrate the system using ambient air at 20.9% O₂.
- Power on the flowmeters and pressure display and check for leaks at low pressure.
- Set the 3-way valves to a series configuration.
- Set the feed pressure with the regulator to a target.
- Adjust the tube effluent flowrate using the valve to target levels. Re-tune pressure if needed.
- Wait until the readings stabilize.
- Record the tube outlet flow, O₂, and shell outlet flow for each condition.
- Repeat the full matrix of pressure x flow for each configuration.
- Perform triplicate at the center point to estimate experimental error.
- Switch the valves to parallel and repeat the run matrix.
- Shut down by closing the cylinder, venting pressure to zero, powering off instruments and restoring valves to safe positions.

In this experiment, the independent variables are the tube-side operating pressure, effluent flowrate, and module configuration. The dependent variables are the measured O₂ mole fractions and outlet flowrates on both the tube and shell streams. A mid-point operating condition is performed in triplicate to quantify repeatability and to produce a 95% confidence interval via pooled variance. Goodness of fit for the solution-diffusion linear transport model is evaluated by regression of permeate molar flux against driving force Δp , reporting R² and testing whether the intercept is statistically indistinguishable from zero. The resulting data will be correlated with the

solution-diffusion membrane model and trends will be compared across configurations to evaluate pressure and flow dependence of observed selectivity and flux.

Project Safety Evaluation (10 pts)

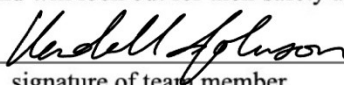
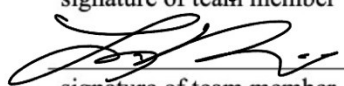
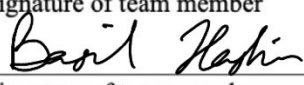
The experiment uses compressed air at a high pressure which is regulated down to controlled feed pressures before entering the membrane modules. The main hazards arise from stored energy in the compressed gas system, potential over-pressurization of the membrane shell side, localized oxygen-enrichment of the permeate stream, and electrical hazards from powered instrumentation. Mis-positioning of the 3-way valves can isolate and pressurize the shell side above its intended ~1 atm operating pressure. Oxygen in the permeate can be enriched relative to feed air, this is not directly toxic but increases flammability of combustible materials near an ignition source, which should be avoided in the area. All power components are connected through a GFCI-protected and properly grounded system, which reduces but doesn't eliminate the possibility of electrical shock.

Safety controls are built into the apparatus and the procedure. A two-stage regulator limits the cylinder discharge pressure and all operating adjustments should be made slowly – pressure should be established before flow is set. PPE should include eye protection, cylinders should remain secured and upright, and lines should be checked for leaks before operation. No toxic chemicals or liquid reagents will be handled in this experiment so now chemical waste is generated. Shutdown will consist of closing the cylinder, venting to zero pressure, backing off the regulator, de-energizing the equipment, and returning all valves to safe positions. These controls will maintain the experiment within a safe operating window and reduce the likelihood and severity of pressure, mechanical, flammability, and electrical hazards.

Figure 2: Safety evaluation chart.

	<i>Yes/No</i>	
Potential electrical hazards?	N	If yes, identify conditions:
Potential mechanical hazards?	N	If yes, identify conditions:
Potential pressure hazards?	Y	If yes, identify conditions: Compressed air (300psi)
Potential temperature hazards?	N	If yes, identify conditions:
Hazardous/toxic chemicals involved?	N	If yes, hazards involved:
Gloves required?	N	If yes, specify type:
MSDS reviewed by all team members?		List MSDS sheets reviewed:
<i>Process Parameters</i>	<i>Max Expected</i>	<i>List location and possible hazards and precautions</i>
Temperature (°C)	RT	
Pressure (psia)	300	Tank of compressed air, chained to table & regulated & pressure release valve
<i>Safety Controls</i>	<i>yes/no</i>	<i>If yes, Provide description</i>
	Y	Pressure regulator, pressure release valves

I have read relevant background material for the Unit Operations Laboratory entitled:
 “ Membrane Air Separation ” and
 understand the hazards associated with conducting this experiment. I have planned out my
 experimental work in accordance to standards and acceptable safety practices and will conduct all
 of my experimental work in a careful and safe manner. I will also be aware of my surroundings,
 my group members, and other lab students, and will look out for their safety as well.

	10/29/25
signature of team member	date
	10/29/25
signature of team member	date
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signature of team member	date